

Fundamentals of Astronomy

Stellar Atmosphere: Radiative transfer equation

Gino Tosti

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The transport of radiation I

We have seen that if a beam of radiation traverses a layer of matter, the intensity added to the beam by spontaneous emission is:

$$dl_{\nu}(s) = j_{\nu}(s)ds$$

where j_{ν} is the spontaneous monochromatic emission coefficient, defined as the energy emitted per unit of time per unit solid angle per unit volume in the frequency range. The amount of absorption that the beam of radiation experiences in passing through a layer of matter of thickness ds is expressed by a coefficient α_{ν} (having unit of m^{-1}) in the equation:

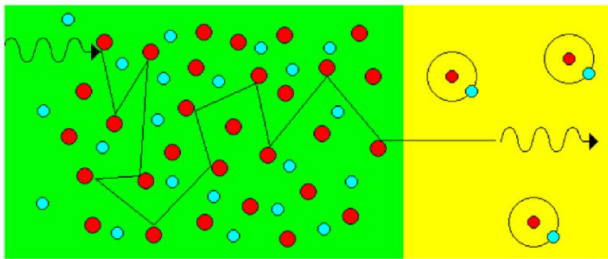
$$dl_{\nu}(s) = -\alpha_{\nu}l_{\nu}(s)ds$$

In the theory of stellar atmospheres we use the absorption for kg of material. If ρ is the density of the matter, we have $\alpha_{\nu} = \kappa_{\nu}\rho$, then:

$$dl_{\nu}(s) = -\kappa_{\nu}\rho l_{\nu}(s)ds$$

κ_{ν} (m^2/kg) is usually called **opacity** and includes both absorption and scattering.

The transport of radiation II



The relation between α_ν and the mean free path of photons ℓ_ν

Let consider the matter crossed by radiation consisting of N particle in volume V each having a radius $d/2$. We want to find an expression for the average distance traveled by a particle between two collision. Viewed head on, each particle appears as a disc with area $\pi(d/2)^2$, then the effective cross-sectional area (σ) of the particle is $\sigma = \pi d^2$. If a particle travels a distance ℓ , it will sweep out a volume $\pi d^2 \ell$ collide

The transport of radiation III

with any other particle whose centre lies within this volume. The **mean free path** is defined to be the average distance travelled by the particle between each collision:

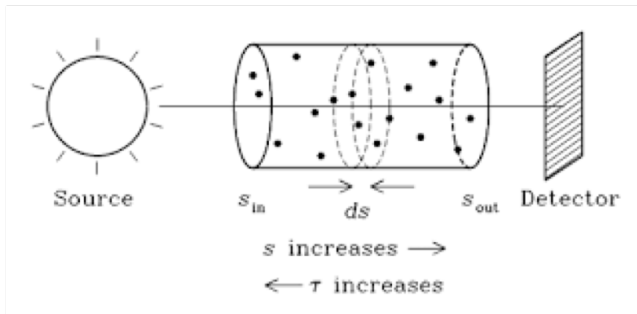
$$\ell = \frac{V}{N\sigma} = \frac{1}{n\sigma}$$

where $n = N/V$ is the particle density.

If we express the interaction of photon with matter through the appropriate cross-section σ_ν , we have that

$$\alpha_\nu = n\sigma_\nu = \frac{1}{\ell_\nu}$$

The transport of radiation IV



Monochromatic transport equation along a ray

To express the changes in the monochromatic intensity of the radiation along a path ds :

$$\frac{dl_{\nu}(s)}{ds} = -\alpha_{\nu}(s)l_{\nu}(s) + j_{\nu}(s)$$

The transport of radiation V

with s measured along the beam in the propagation direction. This equation says that photons do not decay spontaneously so that the intensity along a ray does not change unless photons are added to the beam or taken from it. In the vacuum $I_\nu/ds = 0$.

The Source function

We define as Source function $S_\nu(s)$ as:

$$S_\nu = \frac{j_\nu}{\alpha_\nu} \rightarrow j_\nu = \alpha_\nu S_\nu$$

we have:

$$\frac{dI_\nu(s)}{\alpha_\nu(s)ds} = S_\nu(s) - I_\nu(s)$$

The formation of the stellar spectra is then described in terms of **the parameter α_ν , which gives the atomic particle properties** (such as the gas composition, the degree of ionization and excitation, the probability of spectral line transitions

The transport of radiation VI

at some frequency, the nature and amount of line broadening) and which sets the transparency of the atmosphere (the depth above which the gas is sufficiently transparent that photons may escape towards the observer), and **the parameter S_ν which describe the thermodynamic state of the matter** (particles and photons).

The transport of radiation VII

The optical depth

It is convenient to introduce a dimensionless quantity called **optical depth**:

$$\tau_\nu = \int_0^L \alpha_\nu(s) ds$$

If α_ν is independent from s ,

$$\tau_\nu = L\alpha_\nu = Ln\sigma_\nu \rightarrow \tau_\nu = \frac{L}{\ell_\nu}$$

Then τ give an estimation of the number of mean free path of the photons in the matter. In general after each interaction with particles the photon will not proceed in the original direction, but will perform a "random walk" and it is found that τ_ν^2 gives the number of collision that will have a photon before to exit from a region having linear size L .

If $\tau_\nu = 1$ then $L = \ell_\nu$ then the photon will cross the distance L without interacting with the matter.

The transport of radiation VIII

In terms of the monochromatic optical thickness τ_ν the transport equation became:
($d\tau_\nu = \alpha_\nu ds$):

$$\frac{dl_\nu(\tau_\nu)}{d\tau_\nu} = S_\nu(\tau_\nu) - l_\nu(\tau_\nu)$$

The formation of the absorption and emission lines I

To know the change of the monochromatic intensity $I_\nu(\tau_\nu)$ along a path s we have to integrate the radiative transport equation. The formal result, when we integrate the equation between $\tau_\nu = 0$ when $I_\nu = I_\nu(0)$ and any τ_ν , is:

$$I_\nu(\tau_\nu) = I_\nu(0)e^{-\tau_\nu} + \int_0^{\tau_\nu} S_\nu(t_\nu)e^{-(\tau_\nu-t_\nu)} dt_\nu$$

If we consider an **homogeneous Medium** the source function S_ν is the same at any location, then we can solve the above integral and we obtain:

$$I_\nu(L) = I_\nu(0)e^{-\tau_\nu(L)} + S_\nu(1 - e^{-\tau_\nu(L)})$$

where L is the linear size of the region crossed by the radiation. We can analyze some cases.

The formation of the absorption and emission lines II

$I_\nu(0) = 0$: This correspond to the case of a volume filled with hot gas, no external light is irradiating it. We have

$$I_\nu(L) = S_\nu(1 - e^{-\tau_\nu(L)})$$

We can discuss two cases:

$\tau_\nu(L) \ll 1$: **optically thin case** : in this case we can expand the exponential function in a Taylor series:

$$e^{-\tau_\nu(L)} \simeq 1 - \tau_\nu(L)$$

then:

$$I_\nu(L) = \tau_\nu(L)S_\nu$$

if α_ν and j_ν are constant along s , we have:

The formation of the absorption and emission lines III

$$I_{\nu}(L) = L\alpha_{\nu}S_{\nu} = Lj_{\nu}$$

Then the emerging radiation will follow the behavior of j_{ν} , will be higher where j_{ν} is higher, this usually happen at frequencies corresponding to atomic transitions. In this case we will observe **emission lines**. The **solar corona** corresponds to this case.

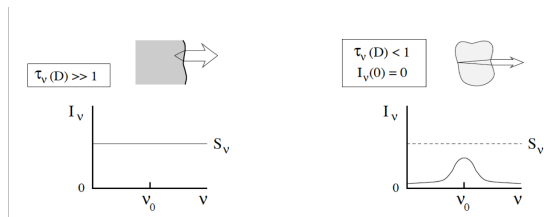
The formation of the absorption and emission lines IV

$\tau_\nu(L) \gg 1$: optically thick case : In this case $e^{-\tau_\nu(L)} \rightarrow 0$
then:

$$I_\nu(L) = S_\nu$$

The emitted radiation is independent of α_ν ,
and no emission lines will be observed. This is
the case of the blackbody when $\tau_\nu \rightarrow \infty$. In
this case $S_\nu = B_\nu$.

The formation of the absorption and emission lines V



$I_\nu(0) \neq 0$ and $\tau_\nu(L) \ll 1$: In this case

$$I_\nu(L) = I_\nu(0) + \tau_\nu(L)(S_\nu - I_\nu(0))$$

If $S_\nu < I_\nu(0)$ then:

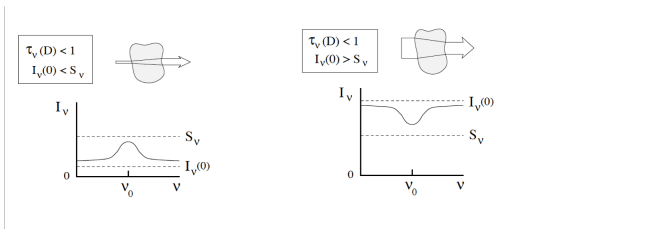
$$I_\nu(L) = I_\nu(0) - \tau_\nu(L)(I_\nu(0) - S_\nu)$$

The formation of the absorption and emission lines VI

In this case there is more intensity missing at those frequencies for which α_ν is higher. In this case we will observe **absorption lines** on the background intensity $I_\nu(0)$. If $S_\nu > I_\nu(0)$ then:

$$I_\nu(L) = I_\nu(0) + \tau_\nu(L)(S_\nu - I_\nu(0))$$

the last term is positive and we will see **emission lines** on top of the background intensity $I_\nu(0)$.



The formation of the absorption and emission lines VII

$I_\nu(0) \neq 0$ and $\tau_\nu(L) \gg 1$: In this case, independently from the value of $I_\nu(0)$:

$$I_\nu(L) = S_\nu$$

The original intensity is totally absorbed along the path s .

In summary, for an homogeneous object:

- No lines emerge when the object is optically thick;
- When it is optically thin, **emission lines** emerge when $I_\nu(0) = 0$ or when it is illuminated with $I_\nu(0) < S_\nu$;
- Absorption lines emerge only when the object is optically thin and $I_\nu(0) > S_\nu$.

The formation of the absorption and emission lines VIII

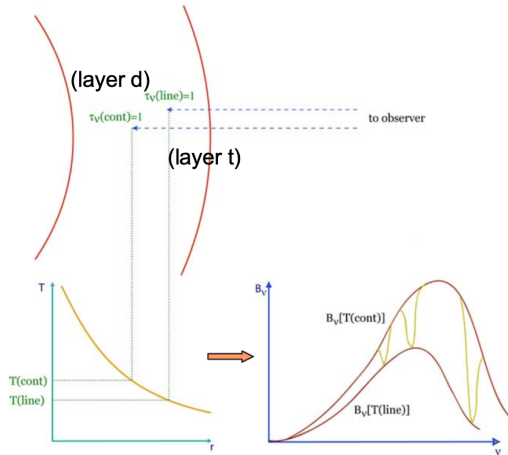
Spectral Line formation in Stars

The stars show absorption lines this means that the $I_\nu(0)$ coming from the deeper layers of the atmosphere is larger than the source function S_ν of the top layers.

If we assume that $S_\nu = B_\nu(T)$ then the source function increases with increasing temperature. We divide the star in two layers a deeper optically thick layer (d) and a top (t) optically thin reversing layers. If $I_\nu(0) = B_{\nu,d}(T_d)$ is the input radiation in the top layer then, because we see absorption lines, the source function $B_{\nu,t}(T_t)$ of the top layer have to be such that:

$$B_{\nu,d}(T_d) > B_{\nu,t}(T_t) \Rightarrow T_d > T_t$$

The formation of the absorption and emission lines IX



The formation of the absorption and emission lines X

As a rough approximation we can consider the stellar spectrum as consisting of two parts. The first is the contribution from the blackbody at the base of the atmosphere $I_\nu = B_\nu(T_d)$, the **spectral continuum**. Subtracted from this is a contribution from the higher layers at the locations of strongly-absorbing lines $I_\nu \simeq B_\nu(T_d) - \alpha_\nu L(B_\nu(T_d) - B_\nu(T_t))$.

In the transport equation the absorption and emission coefficients were defined phenomenologically. To be able to write their actual expressions we have to go to microscopic physics, that is, we have to describe all contributions from microscopic processes that give rise to an absorption or emission of photons.